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# Proof that can be checked.

## Topological structural certification for semiconductor design

Topos Structural Certificate v1 computes a precisely delimited mathematical property of a supplied digital-hardware structural model. Its signed results are independently recomputable.

**AVAILABLE AS A SCOPED B2B ASSESSMENT**

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# The verification gap

## Early structural evidence, rather than another heuristic alarm

Engineering teams work with huge design spaces. A topology-based measurement can identify structural features missed by graph-cycle counts, with explicit mathematical assumptions and auditable evidence.

**We certify model properties, not functional correctness, deadlock freedom, PPA improvement, or regulatory compliance.**

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# A precise and honest product

## Certified kernel v1 / GF(2) / H1

Input: supported normalized structural JSON or Yosys JSON. Output: a design-bound machine-readable certificate containing the first Betti number, a witness when nonzero, cryptographic provenance and an Ed25519 signature.

**Verification recomputes the model and mathematical result. A signature alone is not enough.**

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# Technical evidence, not speculation

## Executable kernel / test corpus / verifiable output

The frozen certificate kernel passed canonical release tests and a complete GitHub Actions acceptance flow on 24 October 2026. Synthetic examples with beta1 = 0 and beta1 = 1 were issued and verified using disposable QA credentials.

**Commercial signing identity and customer-data processing controls are commissioned separately; QA certificates are not production certificates.**

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# Go to market

**Start narrow. Learn with paid design-team pilots.**

Initial customers: semiconductor design teams, EDA vendors and hardware engineering groups. Model: fixed-scope B2B assessment per design, followed by repeat verification, integrations and enterprise licensing if validated.

**Early market presence: Spain / EU. No paying product customers claimed.**

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# Market opportunity

## A specialist wedge into electronic design automation

Global IC physical-design and verification revenue in Q2 2026 was USD 818.3 million (SEMI / ESD Alliance). Annualised at the ECB 23 October 2026 reference rate, that is approximately EUR 2.92 billion. The directly addressable market for v1 is narrower.

**This is an adjacent software category proxy, not a bottom-up estimate of v1 customer spending.**

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# Roadmap and investment

## Demonstrate industrial value before claiming physical savings

0-6 months: first paid B2B engagements, customer verification and secure intake. 6-12 months: controlled comparisons with graph-only baselines and measurements of PPA. 12-24 months: repeatable integrations and scalable distribution.

**Fundraising objective: EUR 750,000 pre-seed (target, not funds raised).**

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# Company and contact

**Founded in Spain in July 2025 / digital semiconductor infrastructure**

ToposCircuitry, S.L. develops formal structural evidence tooling for digital hardware. Founder and managing director: Dr Andreu Ballus Santacana. Patent application pending; independent evaluation of computational performance under way.

**Discuss a scoped pilot: andreuballus@toposcircuitry.com**